

Poster: GEM-PLI Side Channel in GPON Inferring Encrypted Video Streaming from Access-Layer Length Traces

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Abstract—Gigabit Passive Optical Networks (GPON) widely protect user traffic by encrypting downstream payloads at the GPON Encapsulation Method (GEM) layer. However, encryption does not cover all access-layer metadata. In this work-in-progress poster, we provide the first evidence that the plaintext Payload Length Indicator (PLI) in GEM headers can form a practical privacy side channel for video streaming inference. We consider an attacker co-located under the same optical splitter who passively observes downstream GPON transmissions without decrypting payloads. Due to GPON’s downstream broadcast and multiplexing, the attacker can obtain a high-rate time series of GEM payload lengths aligned with access-layer framing, which preserves the structural characteristics of HTTP adaptive streaming (HAS), such as segment-driven bursts and bitrate-dependent size patterns. To study this leakage systematically, we build a GPON downstream trace generator that maps legitimate encrypted video traffic (e.g., YouTube/Netflix sessions) into GEM frames and aggregates them into 125- μ s GPON transmission frames, enabling controlled evaluation of observability at the GEM layer. Using derived PLI sequences only, we demonstrate preliminary separability between streaming and non-streaming activity, and we outline how adaptive bitrate transitions can be detected from length dynamics. Our findings highlight that payload encryption alone may be insufficient to protect behavioral privacy in access networks, motivating mitigations such as padding/shaping and privacy-aware handling of access-layer metadata.

I. INTRODUCTION

End-to-end encryption has become the default for modern video services, yet encryption does not eliminate side channels. Prior work has shown that encrypted traffic may still leak user behavior through patterns in packet size and timing, especially for HTTP adaptive streaming (HAS), where content is delivered in short segments whose sizes and delivery dynamics depend on codec, bitrate, and player behavior. However, most video identification studies observe traffic at the IP/Ethernet layer, where signals are often distorted by host-side effects such as segmentation, retransmissions, congestion control, and operating-system scheduling. These artifacts can blur segment

boundaries and require longer observation windows or heavier training to achieve reliable inference.

In this poster, we explore an under-studied observability surface: the GPON access layer. GPON encrypts user payloads at the GPON Encapsulation Method (GEM) layer, but GEM headers necessarily retain framing metadata in plaintext for real-time multiplexing. We focus on the Payload Length Indicator (PLI) in GEM headers, which exposes the encrypted payload length at high rate. Because downstream GPON traffic is broadcast over the splitter, a co-located observer may obtain an access-layer time series aligned with GPON’s fixed 125 μ s transmission framing, potentially preserving HAS-induced structure more clearly than IP-layer traces. Our goal is to provide early evidence of this privacy leakage channel and outline practical defenses for future access networks.

a) *Validation protocol.*: To validate that the “access-layer trace is cleaner” hypothesis is testable (not merely asserted), we (i) define a measurable notion of “cleanliness,” such as segment-boundary recoverability or burst periodicity, (ii) compare the same session observed at two vantage points—IP-layer packet sizes vs. access-layer (simulated) PLI/125 μ s frame occupancy—using the same preprocessing and window length, and (iii) report at least one quantitative metric (e.g., autocorrelation peak strength, burstiness index, or boundary detection F1) rather than relying on qualitative plots alone.

II. THREAT MODEL

We consider a passive attacker located under the same optical splitter as a victim subscriber in a GPON access network. The attacker does not decrypt payloads and does not require control of the OLT or operator infrastructure. Instead, the attacker aims to obtain access-layer metadata that remains observable despite payload encryption. In an idealized “Stealth Rogue ONU” model, the attacker’s device behaves as a standards-compliant ONU at higher layers but can observe downstream GPON transmissions beyond its intended logical isolation boundary. Concretely, the attacker extracts GEM header fields—primarily PLI—over time, producing a length time series $\{L_t\}$ that reflects multiplexed downstream traffic.

We focus on three inference goals, ordered by difficulty: (1) *session detection* (is the victim streaming now?), (2) *quality tracking* (detecting adaptive bitrate transitions), and (3) *content fingerprinting* (matching a short window to a known catalog).

This poster emphasizes (1) and (2) as near-term demonstrable leakage, and treats (3) as an extensible direction requiring larger datasets and stronger controls. Ethically, we assume all measurements are conducted on a controlled testbed with the experimenter’s own traffic, and we analyze metadata only.

a) Validation protocol.: To validate that the threat model is neither overpowered nor underspecified, we (i) explicitly bound capabilities (no payload decryption; metadata-only; controlled testbed), (ii) include a negative control showing that inference fails when PLI is padded/shaped (simulated defense), and (iii) state what would invalidate the model (e.g., if actual OLT/ONU implementations strictly prevent any non-destined downstream metadata observation, then the attacker reduces to an IP-layer observer and our contribution becomes a baseline comparison rather than a new access-layer leakage channel).

III. TRACE GENERATION (125 μ s)

To study PLI leakage systematically without relying on proprietary GPON capture hardware, we build a GPON downstream trace generator that transforms legitimate encrypted video traffic into access-layer observables. Starting from packet captures (PCAP) of video sessions (e.g., YouTube and Netflix), we map Ethernet frames into GEM payloads, fragmenting frames when necessary to satisfy the GEM payload length bound (PLI is 12-bit). For each GEM unit, we construct a header containing PLI and other demultiplexing fields while keeping the payload encrypted using AES-CTR in a model consistent with GPON’s “payload encrypted, header plaintext” design. The resulting GEM stream is then packed into fixed-length 125 μ s transmission frames, representing the GPON downstream framing clock. From this packed stream we derive the observer’s signal in two equivalent forms: (a) the GEM-level sequence of PLI values, and (b) the per-125 μ s “frame occupancy” series (bytes of GEM payload transmitted per GPON frame), which reflects scheduling-aligned timing.

This simulator enables controlled experiments over window size, background load, and multiplexing. It also supports ablations (e.g., padding, reshaping, and noise injection) to estimate how robust the leakage is under practical defenses and realistic cross-traffic.

a) Validation protocol.: To validate that the generator preserves HAS-relevant structure while not inventing artifacts, we will (i) sanity-check that the total bytes emitted by GEM payloads match the input traffic volume (up to known overhead such as headers), (ii) verify fragmentation correctness by confirming that concatenating GEM payloads reconstructs the original frame stream (before encryption), (iii) confirm that the 125 μ s packing obeys the per-frame byte budget implied by the configured line rate, and (iv) run round-trip tests where the same session is generated multiple times to ensure determinism (or controlled randomness) so that observed variance can be attributed to experimental factors rather than simulator instability.

IV. PRELIMINARY RESULTS

Our preliminary analysis suggests that GEM-PLI-derived signals capture structured bursts consistent with segment-

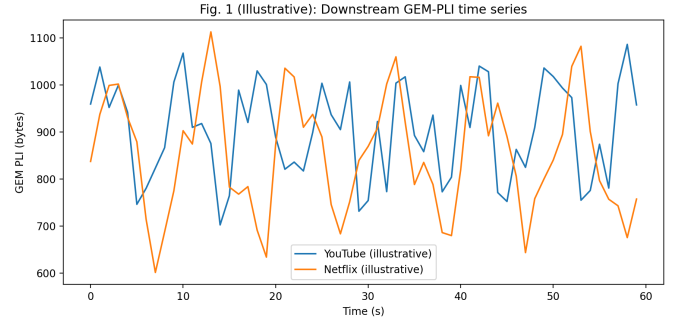


Fig. 1. **Fig. 3 (one-line).** Example GEM-PLI (or 125 μ s frame occupancy) time series for YouTube vs. Netflix sessions, highlighting segment-driven burst structure and candidate ABR transition points.

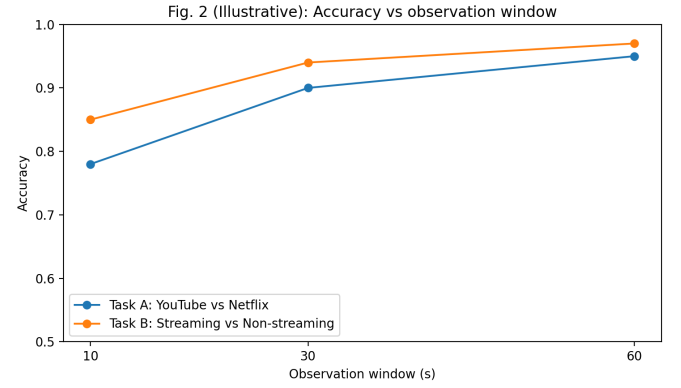


Fig. 2. **Fig. 4 (one-line).** Classification accuracy vs. observation window length (e.g., 10/30/60 s) for streaming-session detection and preliminary ABR-change detection.

based delivery, even when payloads are encrypted. In controlled sessions, PLI time series show recurring burst patterns whose intensity and dispersion vary with service and playback conditions. Using simple features over short windows (e.g., windowed sums, burstiness measures, and coarse histograms of lengths), we observe separability between streaming and non-streaming activity, supporting the feasibility of session detection with minimal assumptions. We further outline a practical approach to detect adaptive bitrate transitions by tracking shifts in the distribution of observed lengths and inter-burst dynamics, which are expected to change when the player switches representation.

a) Validation protocol.: To validate that these results are not overfitting or dataset leakage, we will (i) use session-level splits (train/test split by video/session, not by window) to prevent overlapping segments across splits, (ii) report confidence intervals via bootstrapping over sessions, (iii) include baselines (random guess, IP-layer-only features, and a simple threshold detector) to contextualize gains, and (iv) perform ablations over background traffic and multiplexing intensity to show whether the signal persists when realistic noise is introduced.

V. IMPACT & MITIGATION

If access-layer metadata such as GEM-PLI enables reliable inference of video usage, then payload encryption alone is insufficient to protect behavioral privacy in optical access networks. This leakage is concerning because it arises below the IP layer and can be available at high rate, aligned with access-layer framing. The impact ranges from coarse inferences (whether a household is streaming) to finer-grained behavioral profiling (quality changes that correlate with device type, network conditions, or user choices), and potentially content fingerprinting with larger reference datasets.

We discuss two mitigation directions. First, *padding or traffic shaping* at the GEM or scheduling level can reduce information leakage by flattening length dynamics, at the cost of bandwidth overhead and latency. Second, *privacy-aware access-layer design* could limit exposure of fine-grained metadata, for example by aggregating, quantizing, or obfuscating length indicators where feasible. Our simulator supports evaluating the privacy-overhead trade-off by varying padding policies and measuring their effect on both inference accuracy and bandwidth increase.

a) Validation protocol.: To validate mitigation claims, we will (i) implement at least two defense variants (e.g., fixed-size padding per GEM vs. per-125 μ s frame shaping), (ii) report the privacy-overhead curve (accuracy drop vs. bandwidth overhead), (iii) test defenses under multiple services and playback conditions, and (iv) document feasibility constraints (which defense requires endpoint changes vs. could be deployed at access equipment).

VI. CONCLUSION

We present early evidence that plaintext GEM-PLI metadata in GPON may enable streaming session and quality inference without payload decryption, motivating access-layer padding and shaping defenses.

ACKNOWLEDGMENT

The authors would like to thank...

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